8(a) One end of a string is secured to the ceiling and a metal ball of mass 50 g is tied to its other end. The length of the string is 1.50 m. The ball is initially at rest in the vertical position. The ball is raised through a vertical height of 70 cm (see **Fig. 8.1** below). When the ball is released, it describes a circular arc as it passes through the vertical position.

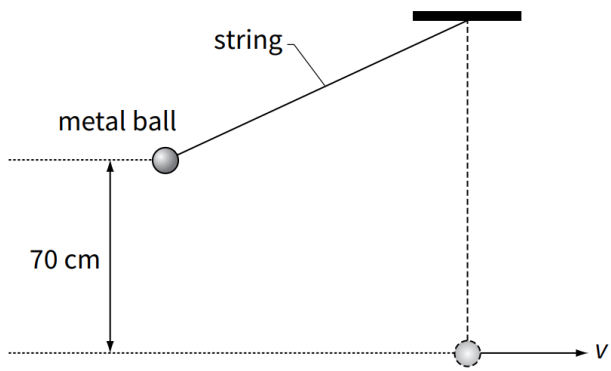

Fig. 8.1

(i) Ignoring the effects of air resistance, show that the speed v of the ball is 3.7 m s^{-1} as it passes through the vertical position. [1]

(ii) Draw a labelled force diagram to show all the forces acting on the ball when the string is in the vertical position. [2]

(iii) Calculate the centripetal force on the ball when it is in the vertical position. [2]

(iv) Explain why the tension in the string when it is vertical is not equal to the weight of the ball. [2]

(v) Calculate the tension in the string when it is vertical. [2]

(b) **Fig. 8.2** shows the dwarf planet, Pluto, and its moon, Charon. These can be considered to be a double planetary system orbiting each other about their joint centre of mass. To simplify the situation, assume that Pluto has a circular orbit of radius r_P and Charon has a circular orbit r_C. The distance between the centres of mass of Pluto and Charon is 1.96×10^4 km.

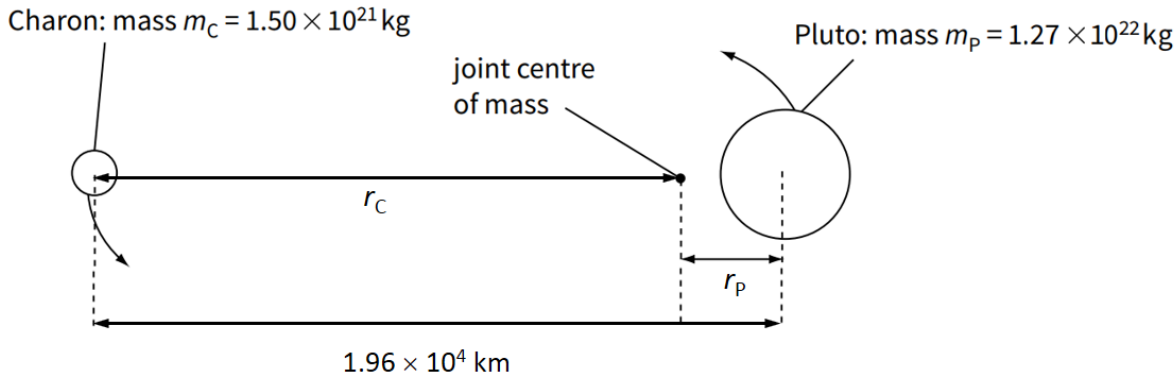

Fig. 8.2

(i) Explain what is meant by the gravitational field strength at a point. [1]

(ii) Calculate the gravitational field strength at the centre of Charon due to Pluto alone, and state its direction. [3]

(iii) Both Pluto and Charon have the same angular speed about their joint centre of mass. Show that the ratio of the masses is given by the expression: [3]

$$\frac{m_P}{m_C} = \frac{r_C}{r_P}.$$

(iv) Calculate the radius r_C of Charon's orbit. [2]

(v) Calculate Charon's orbital period. [2]